

Land Use/Land Cover Transformation and Forecasting in Gorakhpur City Using GIS and Deep Learning Approaches

A Project Report

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ABSTRACT

Rapid urbanization in medium-sized Indian cities has significantly altered land use/land cover (LULC) patterns, leading to environmental stress and planning challenges. Gorakhpur City, Uttar Pradesh, has experienced substantial spatial transformation over the past two decades due to population growth and infrastructure expansion. This study aims to analyze multi-temporal LULC transformations from 2005 to 2025 and forecast future land use scenarios for 2035, 2045, and 2055 using GIS and deep learning approaches. Multi-date satellite datasets for 2005, 2015, and 2025 were processed and classified into five major categories: agriculture land, built-up area, forest, wasteland, and water bodies. Supervised classification techniques were employed within a GIS framework, followed by change detection analysis to quantify spatial transitions. To predict future LULC dynamics, Artificial Neural Networks (ANN) and Multilayer Perceptron (MLP) models were implemented using historical land transformation patterns. The results reveal a pronounced expansion of built-up areas between 2005 and 2025, primarily at the expense of agricultural land. Urban growth exhibited radial and corridor-based expansion patterns, indicating peri-urban conversion and densification of the urban core. Agricultural land showed a consistent decline, while forest and water bodies experienced localized fluctuations. Forecast results for 2035–2055 suggest continued urban sprawl, significant reduction in agricultural zones, and increasing pressure on environmentally sensitive areas. Among the models, the MLP approach demonstrated improved spatial consistency and prediction accuracy compared to conventional ANN implementation. The findings highlight the urgent need for sustainable urban planning strategies, protection of agricultural buffers, and conservation of water resources. Deep learning-integrated GIS modeling provides a reliable decision-support tool for managing rapid urban expansion in growing cities like Gorakhpur.

Keywords: Land Use/Land Cover (LULC) Change, Urban Expansion Modeling, Artificial Neural Networks (ANN), Multilayer Perceptron (MLP), GIS-Based Urban Forecasting

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1.1 Background

Land Use and Land Cover (LULC) represent the spatial distribution of natural and anthropogenic features on the Earth's surface and play a critical role in understanding environmental dynamics, ecological balance, and human–environment interactions. Over the past few decades, rapid population growth, urbanization, and economic development have significantly altered land use patterns across the globe, particularly in developing countries such as India (Lambin et al., 2003).

Urbanization is one of the most prominent drivers of LULC transformation. The expansion of cities leads to the conversion of agricultural land, forests, wetlands, and other natural landscapes into built-up areas. This transformation not only affects biodiversity and ecosystem services but also alters hydrological processes, increases surface runoff, and intensifies the risk of natural disasters such as floods (Seto et al., 2012). In flood-prone regions, unplanned urban expansion further exacerbates vulnerability by reducing infiltration capacity and encroaching upon natural drainage systems.

The monitoring and analysis of LULC changes are therefore essential for sustainable urban planning, environmental management, and disaster mitigation. Traditional ground-based methods of land use mapping are often time-consuming, costly, and limited in spatial coverage. In contrast, Remote Sensing provides a synoptic, repetitive, and cost-effective means of observing the Earth's surface over time (Campbell & Wynne, 2011). Satellite imagery enables multi-temporal analysis, allowing researchers to detect and quantify land use changes accurately.

Geographic Information Systems (GIS) complement remote sensing by providing tools for spatial analysis, data integration, and visualization. GIS facilitates the identification of spatial patterns and relationships, making it an indispensable tool for LULC studies (Burrough & McDonnell, 1998). Together, remote sensing and GIS form a powerful framework for analyzing past and present land use dynamics.

In recent years, advancements in Artificial Intelligence (AI) and Deep Learning have revolutionized spatial modeling and prediction. Traditional statistical and rule-based models

often fail to capture the complex, non-linear relationships inherent in land use systems. Deep learning models such as Artificial Neural Networks (ANN) and Multilayer Perceptron (MLP) have shown significant potential in modeling such complexities (Goodfellow et al., 2016). These models learn patterns from historical data and can predict future scenarios with improved accuracy.

The integration of GIS with deep learning techniques enables not only the analysis of historical LULC changes but also the forecasting of future land use patterns. Such predictive capabilities are crucial for long-term urban planning, resource management, and environmental sustainability.

1.2 Study Area Overview

Gorakhpur city, located in the eastern part of Uttar Pradesh, India, is an important urban center situated in the middle Gangetic plain. Geographically, the city lies between approximately 26°45'N to 26°51'N latitude and 83°20'E to 83°30'E longitude. The study area comprises the Gorakhpur municipal region within the administrative boundaries of Gorakhpur district and tehsil.

The geographical location and administrative extent of the study area are illustrated in Figure 1.1, which presents a multi-scale representation of Gorakhpur city within Uttar Pradesh, along with its district and tehsil boundaries. This figure provides a clear spatial context of the study area and helps in understanding its regional setting.

The region is characterized by predominantly flat terrain and is part of the fertile alluvial plains formed by the Ganga River system. The presence of rich alluvial soil makes the area highly suitable for agriculture. The city is located near major river systems, particularly the Rapti River, which plays a significant role in shaping the geomorphology and hydrology of the region.

Climatically, Gorakhpur experiences a humid subtropical climate, characterized by hot summers, moderate winters, and intense monsoon rainfall. The region receives substantial precipitation during the monsoon season, which often leads to waterlogging and flooding. The proximity to the Rapti River and its tributaries further increases the flood vulnerability of the area. Historical records and previous studies indicate that Gorakhpur frequently experiences flood events due to river overflow and inadequate drainage capacity (Sinha et

al., 2014).

Over the past two decades, Gorakhpur has undergone rapid urbanization driven by population growth, infrastructural expansion, and socio-economic development. This rapid growth has resulted in significant transformations in land use and land cover patterns. Agricultural lands and open spaces have increasingly been converted into built-up areas, while natural water bodies have been encroached upon or reduced in size.

These land use transformations have critical implications for environmental sustainability and disaster management. The reduction in permeable surfaces and natural drainage systems has increased surface runoff, thereby intensifying the risk of urban flooding. Additionally, the loss of vegetation cover has impacted local microclimatic conditions and ecological balance.

Therefore, a comprehensive analysis of LULC changes in Gorakhpur city, along with future prediction using advanced techniques, is essential for sustainable urban planning, effective resource management, and mitigation of environmental risks.

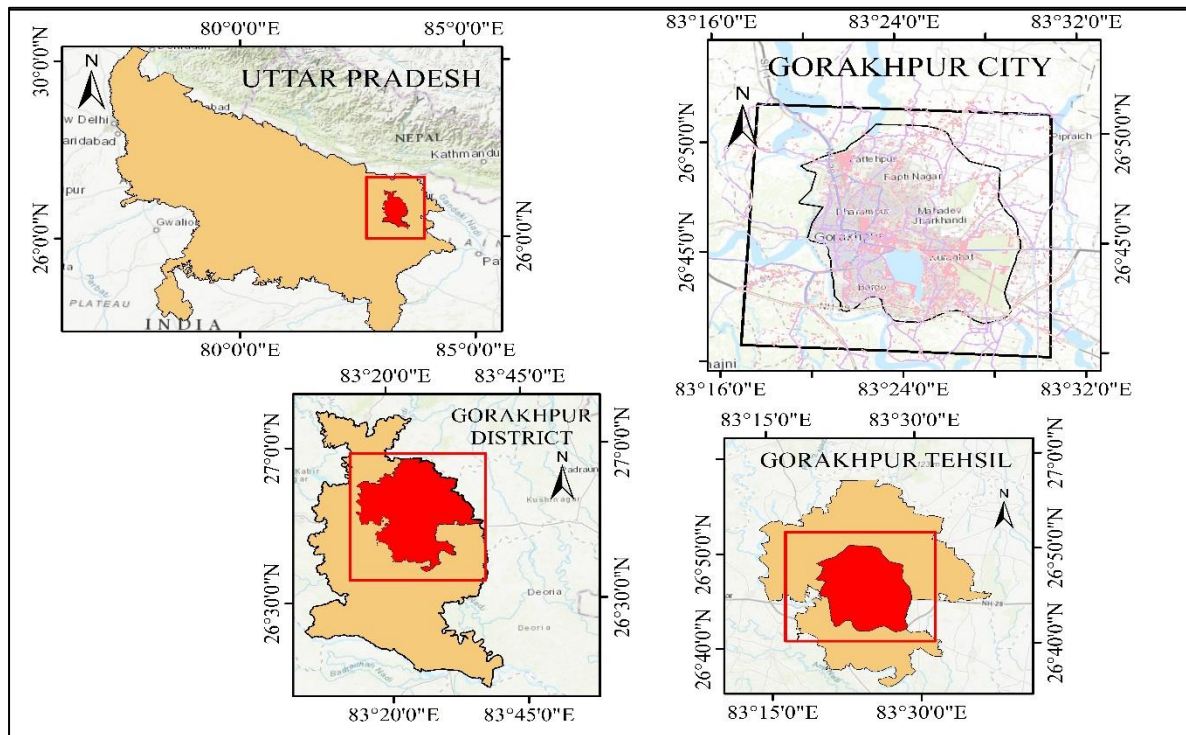


Figure 1. Multi-scale Location Map of the Study Area Showing Gorakhpur City within Uttar Pradesh, Gorakhpur District, and Gorakhpur Tehsil Boundaries

1.3 Problem Statement

Urbanization is one of the most significant drivers of land use and land cover (LULC) transformation, particularly in rapidly developing regions such as eastern Uttar Pradesh. Gorakhpur city has experienced substantial urban expansion over the past two decades due to increasing population pressure, infrastructural development, and economic activities. This rapid and often unplanned growth has led to the conversion of agricultural land, vegetation cover, and open spaces into built-up areas, thereby altering the natural landscape and ecological balance of the region.

Despite these noticeable transformations, there is a lack of comprehensive and systematic studies that integrate both historical LULC analysis and future prediction for Gorakhpur city. Most existing studies are limited to mapping and analyzing past land use changes using remote sensing and GIS techniques, without extending the analysis toward forecasting future scenarios. While these studies provide valuable insights into past trends, they do not offer predictive capabilities necessary for long-term planning and sustainable development.

Traditional LULC prediction models, such as Markov Chain and Cellular Automata (CA), have been widely used for simulating land use changes. However, these models have inherent limitations in capturing complex spatial interactions and non-linear relationships among different land use classes (Clarke et al., 1997). As urban systems are influenced by multiple dynamic and interdependent factors, these conventional approaches often fail to produce highly accurate and reliable predictions, especially in heterogeneous and rapidly evolving urban environments.

In recent years, machine learning and deep learning approaches have emerged as powerful tools for spatial modeling and prediction. Models such as Artificial Neural Networks (ANN) and Multilayer Perceptron (MLP) are capable of learning complex patterns from historical data and modeling non-linear relationships effectively (Goodfellow et al., 2016). However, the application of these advanced techniques for LULC prediction in medium-sized cities like Gorakhpur remains limited, indicating a significant research gap.

Furthermore, Gorakhpur is a flood-prone region located in the floodplains of the Rapti River, where land use changes have a direct impact on hydrological processes and flood vulnerability. The expansion of built-up areas and encroachment on natural drainage systems have increased surface runoff and reduced water infiltration, thereby exacerbating urban

flooding risks (Sinha et al., 2014). In this context, understanding future LULC dynamics becomes critically important for disaster risk reduction, infrastructure planning, and environmental sustainability.

Therefore, there is a pressing need for a comprehensive study that integrates remote sensing, GIS, and deep learning approaches to analyze past LULC changes and predict future land use patterns in Gorakhpur city. Such an approach will not only enhance the accuracy of LULC prediction but also provide valuable insights for policymakers, urban planners, and environmental managers to develop sustainable and resilient urban strategies.

1.4 Aim of the Study

The primary aim of this study is to analyze the spatial and temporal transformations in land use and land cover (LULC) and to forecast future LULC patterns in Gorakhpur city using an integrated approach of Remote Sensing, Geographic Information Systems (GIS), and deep learning techniques. The study seeks to develop a robust predictive framework that can effectively capture complex land use dynamics and provide reliable future scenarios to support sustainable urban planning and environmental management.

1.5 Objectives of the Study

To achieve the above aim, the following specific objectives have been formulated:

1. To generate multi-temporal land use/land cover (LULC) maps for the years 2005, 2015, and 2025 using satellite imagery and supervised classification techniques, thereby establishing a reliable baseline dataset for analysis (Anderson et al., 1976).
2. To analyze spatial and temporal changes in LULC by examining patterns and trends over the study period, including the extent and direction of urban expansion, agricultural decline, and variations in natural land cover (Singh, 1989).
3. To quantify LULC class transitions by calculating area statistics and percentage changes of major land use categories such as agriculture, built-up, forest, wasteland, and water bodies, in order to understand the magnitude and intensity of land transformations.
4. To develop deep learning-based predictive models using Artificial Neural Networks (ANN) and Multilayer Perceptron (MLP) for capturing complex and non-linear

- spatial relationships and predicting future land use patterns (Goodfellow et al., 2016).
5. To forecast future LULC scenarios for the years 2035, 2045, and 2055 based on historical trends and model outputs, providing long-term projections of urban growth and land transformation.
 6. To evaluate the environmental and urban planning implications of LULC changes, particularly in terms of flood risk, loss of agricultural land, and degradation of natural resources, and to support sustainable planning and policy formulation (Seto et al., 2012).

1.6 Significance of the Study

This study holds significant importance in both academic and practical contexts, as it addresses critical challenges associated with land use transformation and urban expansion in a rapidly developing and flood-prone region.

From an academic perspective, the study contributes to the existing body of knowledge on LULC change analysis by integrating traditional geospatial techniques with advanced deep learning approaches. The combined use of Remote Sensing and Geographic Information Systems (GIS) enables accurate mapping and analysis of historical land use changes, while the incorporation of Artificial Neural Networks (ANN) and Multilayer Perceptron (MLP) enhances the capability to model complex, non-linear spatial relationships (Goodfellow et al., 2016). This integration represents a significant advancement over conventional methods and provides a more robust framework for LULC prediction. Furthermore, the study adds to the limited research available on the application of deep learning techniques for medium-sized Indian cities, thereby filling an important research gap.

From a practical perspective, the study offers valuable insights for urban planners, policymakers, and environmental managers. By identifying key trends such as rapid urban expansion, reduction in agricultural land, and encroachment on water bodies, the study provides a scientific basis for informed decision-making. These insights can support the formulation of sustainable land use policies, efficient resource allocation, and improved urban infrastructure planning (Seto et al., 2012).

In addition, Gorakhpur's location in the floodplains of the Rapti River makes it highly vulnerable to flooding. Land use changes, particularly the increase in impervious surfaces

and loss of natural drainage systems, have a direct impact on hydrological processes and flood risk (Sinha et al., 2014). In this context, the study plays a crucial role in disaster risk reduction by identifying areas that are likely to become more vulnerable due to future land use changes.

The predictive outputs generated in this study can be utilized for long-term urban planning and environmental management. They can assist in designing strategies for sustainable urban growth, conservation of natural resources, protection of water bodies, and development of green infrastructure. Ultimately, the study contributes toward enhancing urban resilience and promoting sustainable development in Gorakhpur city.

1.7 Scope of the Study

The present study focuses on analyzing historical land use/land cover (LULC) dynamics and forecasting future land use scenarios for Gorakhpur city using an integrated approach of Remote Sensing, Geographic Information Systems (GIS), and deep learning techniques.

The scope of the study includes a multi-temporal analysis of LULC using satellite imagery for the years 2005, 2015, and 2025, enabling the identification of spatial and temporal changes over two decades. The study involves the classification of land use into major categories such as agriculture, built-up, forest, wasteland, and water bodies using supervised classification techniques. This classification forms the basis for detecting and quantifying changes in different land use classes over time.

Furthermore, the study extends beyond conventional analysis by incorporating deep learning models, specifically Artificial Neural Networks (ANN) and Multilayer Perceptron (MLP), to predict future land use patterns. Based on historical trends and model training, future LULC scenarios are forecasted for the years 2035, 2045, and 2055, providing long-term insights into urban expansion and land transformation.

The study also aims to assess the implications of these changes in terms of environmental sustainability and urban planning, particularly in a flood-prone region like Gorakhpur.

However, certain limitations exist within the scope of the study. The analysis is primarily based on remotely sensed data and does not incorporate socio-economic variables such as population density, economic growth, or policy interventions, which can influence land use dynamics. Additionally, the study does not include detailed hydrological or flood modeling,

although it highlights potential implications related to flood vulnerability. The accuracy of classification and prediction is also dependent on the quality, resolution, and availability of satellite data and training samples.

Despite these limitations, the study provides a comprehensive framework for understanding past, present, and future LULC dynamics using advanced geospatial and computational techniques.

1.8 Organization of the Report

The present report is systematically organized into five chapters to ensure a logical flow of information and comprehensive coverage of the study.

1. Chapter 1: Introduction – Provides an overview of the study, including background, study area description, problem statement, aim and objectives, significance, and scope of the research.
2. Chapter 2: Literature Review – Presents a detailed review of previous studies related to land use/land cover mapping, change detection techniques, and prediction models using machine learning and deep learning approaches. It also identifies the research gaps addressed in the present study.
3. Chapter 3: Data and Methodology – Describes the datasets used, including satellite imagery and ancillary data, along with preprocessing techniques. It further explains the methodology adopted for LULC classification, accuracy assessment, change detection, and the development of ANN and MLP models for prediction.
4. Chapter 4: Results and Discussion – Presents the results of LULC classification for different years, change detection analysis, and predicted future scenarios. It also includes a detailed discussion of spatial patterns, trends, and implications of land use changes.
5. Chapter 5: Conclusion and Recommendations – Summarizes the key findings of the study and provides recommendations for sustainable urban planning and environmental management. It also highlights the limitations of the study and suggests directions for future research.

2.1 Overview of Land Use/Land Cover (LULC) Dynamics

Land Use and Land Cover (LULC) dynamics refer to the spatial and temporal variations in the Earth's surface resulting from both natural processes and human-induced activities. These dynamics are influenced by a range of factors including urbanization, population growth, agricultural expansion, industrialization, and infrastructural development (Lambin et al., 2003). In recent decades, rapid urban growth has emerged as one of the most dominant drivers of LULC change, particularly in developing countries such as India, where increasing population pressure and economic development have accelerated land transformation processes.

The continuous modification of land use patterns has significant implications for environmental systems. Changes in LULC directly affect hydrological processes by altering infiltration rates, evapotranspiration, and surface runoff. The conversion of natural and agricultural land into built-up areas increases the extent of impervious surfaces, which reduces groundwater recharge and enhances runoff, thereby increasing the risk of urban flooding (Seto et al., 2012). In addition, LULC changes contribute to soil degradation, loss of vegetation cover, and decline in biodiversity, ultimately disrupting ecological balance.

Furthermore, the transformation of land use patterns plays a crucial role in influencing climate variability and environmental sustainability. Urban expansion often leads to the development of urban heat islands, changes in local microclimate, and increased greenhouse gas emissions. These impacts are particularly significant in rapidly growing cities where land use changes occur at a faster rate than environmental management measures.

In the context of developing regions, unplanned urbanization and inefficient land management practices further exacerbate environmental challenges. The encroachment of built-up areas into agricultural lands and water bodies not only threatens food security but also reduces the natural capacity of the landscape to manage water and other resources effectively.

Therefore, a comprehensive understanding of LULC dynamics is essential for sustainable

urban planning, environmental conservation, and disaster risk reduction. Accurate mapping, monitoring, and analysis of land use changes provide valuable insights into the patterns, processes, and driving forces of land transformation. These insights are crucial for developing effective policies and strategies aimed at achieving long-term environmental sustainability and resilient urban development.

2.2 Remote Sensing and GIS in LULC Mapping

Remote sensing and Geographic Information Systems (GIS) have emerged as the most effective and widely adopted tools for land use/land cover (LULC) mapping and analysis. Remote sensing provides multi-temporal and synoptic satellite imagery, enabling continuous observation and monitoring of land surface changes over time. These capabilities make it possible to detect spatial patterns, identify land transformation processes, and analyze environmental changes at various scales (Campbell & Wynne, 2011).

Satellite datasets such as Landsat and Sentinel are extensively used in LULC studies due to their free availability, suitable spatial resolution, and long-term temporal coverage (Wulder et al., 2019). Landsat imagery, with its historical archive spanning several decades, allows for long-term change detection analysis, while Sentinel data offers higher spatial resolution and improved revisit frequency, enhancing the accuracy of recent LULC assessments. The combination of these datasets provides a comprehensive framework for analyzing land use dynamics over both short and long time periods.

LULC classification is a fundamental step in remote sensing analysis, where satellite images are categorized into different land use classes based on their spectral characteristics. Among various classification methods, supervised classification techniques are widely preferred due to their higher accuracy and reliability. Maximum Likelihood Classification (MLC) is one of the most commonly used supervised classification methods, as it considers the statistical distribution of training data and assigns each pixel to the class with the highest probability (Anderson et al., 1976). This approach enables the classification of land into categories such as agriculture, built-up areas, forest, wasteland, and water bodies.

In addition to classification, GIS plays a crucial role in managing, analyzing, and visualizing spatial data. It provides a platform for integrating multiple datasets such as LULC maps, Digital Elevation Models (DEM), road networks, drainage systems, and administrative

boundaries. Through spatial analysis techniques such as overlay analysis, buffer analysis, proximity analysis, and change detection, GIS enables a comprehensive understanding of spatial relationships and land use patterns (Burrough & McDonnell, 1998).

Furthermore, GIS facilitates the quantification of land use changes by allowing the calculation of area statistics, transition matrices, and spatial distribution patterns. The integration of remote sensing and GIS thus provides a powerful and efficient framework for LULC mapping, change detection, and environmental analysis.

Overall, the combined use of remote sensing and GIS has significantly enhanced the accuracy, efficiency, and reliability of LULC studies, making them indispensable tools for urban planning, natural resource management, and environmental sustainability assessment.

2.3 LULC Change Detection Techniques

Change detection analysis is a fundamental component of land use/land cover (LULC) studies, as it enables the identification, quantification, and interpretation of changes in land use patterns over time. By comparing multi-temporal datasets, change detection techniques help in understanding the dynamics of land transformation and the underlying driving forces associated with natural processes and human activities (Singh, 1989).

A variety of techniques have been developed for detecting LULC changes, each with its own advantages and limitations. Commonly used methods include image differencing, image ratioing, principal component analysis (PCA), and post-classification comparison. Image differencing involves subtracting pixel values of images acquired at different time periods to highlight areas of change, while PCA transforms multi-temporal data into new components that emphasize variability and change patterns. Although these methods are computationally efficient, they often provide limited information regarding the specific nature and type of land use transitions.

Among the available approaches, post-classification comparison is the most widely used and reliable method for LULC change detection. This method involves independently classifying images from different time periods and then comparing the classified maps to identify transitions between land use categories. One of the key advantages of this approach is that it provides detailed class-wise change information, allowing researchers to quantify how much area has been converted from one land use class to another. Additionally, it reduces the

influence of atmospheric and sensor-related differences between images, thereby improving the accuracy of change detection results.

The results obtained from post-classification comparison are often represented in the form of change matrices, which provide a comprehensive overview of land use transitions between different classes. These matrices are particularly useful for identifying dominant trends such as the conversion of agricultural land into built-up areas or the reduction of forest cover.

Numerous studies have reported significant urban expansion and land transformation across the globe, particularly in developing regions. Rapid urban growth has led to the conversion of agricultural land, vegetation, and open spaces into built-up areas, thereby altering the natural landscape and environmental conditions. According to global assessments, urban land cover is expected to increase substantially in the coming decades due to population growth and economic development (Seto et al., 2012).

Such transformations have critical implications for environmental sustainability and urban resilience. The increase in impervious surfaces contributes to higher surface runoff, reduced groundwater recharge, and increased risk of urban flooding. Additionally, the loss of vegetation and natural land cover affects biodiversity, microclimate, and ecological stability. Therefore, effective change detection techniques play a crucial role in providing accurate and reliable information on land use dynamics, which is essential for sustainable planning, environmental management, and policy formulation.

2.4 Traditional Models for LULC Prediction

Traditional models have played a significant role in the prediction of land use/land cover (LULC) changes and have been widely applied in spatial modeling studies. Among these, Markov Chain and Cellular Automata (CA) models are the most commonly used approaches for simulating future land use patterns. These models are particularly popular due to their simplicity, ease of implementation, and ability to utilize historical land use data for prediction.

The Markov Chain model is a stochastic process that predicts future land use based on transition probabilities derived from past land use changes. It assumes that future states depend only on the current state and not on the sequence of previous states. By analyzing historical LULC maps, transition probability matrices are generated, which indicate the

likelihood of conversion from one land use class to another over time. This method is effective in capturing temporal changes and providing quantitative estimates of land use transitions.

On the other hand, Cellular Automata (CA) models are spatially explicit models that simulate land use changes based on local neighborhood interactions. In CA models, the study area is divided into a grid of cells, and the state of each cell is determined by predefined transition rules influenced by the characteristics of neighboring cells (Clarke et al., 1997). This approach is particularly useful for modeling spatial patterns such as urban expansion, as it considers the influence of surrounding land use conditions.

The integration of Markov Chain and Cellular Automata (CA–Markov) models has been widely adopted in LULC prediction studies. This hybrid approach combines the temporal prediction capability of Markov models with the spatial simulation strength of CA models, resulting in improved prediction performance.

Despite their widespread use, traditional models have several limitations. One of the primary drawbacks is their inability to effectively capture complex, non-linear relationships between different land use classes. These models often rely on predefined transition rules and assumptions, which may not fully represent the dynamic and heterogeneous nature of real-world land use systems. Additionally, they may struggle to incorporate multiple influencing factors such as socio-economic conditions, infrastructure development, and environmental variables.

As urban systems become increasingly complex, the limitations of traditional models become more evident. The rapid pace of urbanization and the influence of multiple interacting factors require more advanced modeling approaches capable of handling non-linearity and high-dimensional data.

Therefore, while traditional models such as Markov Chain and Cellular Automata provide a useful foundation for LULC prediction, there is a growing need for more sophisticated techniques, such as machine learning and deep learning models, to improve prediction accuracy and better represent real-world land use dynamics.

2.5 Machine Learning Approaches in LULC Prediction

Machine learning techniques have gained significant attention in land use/land cover

(LULC) prediction due to their ability to process large datasets, identify hidden patterns, and model complex relationships between multiple variables. Unlike traditional statistical methods, machine learning algorithms do not rely on strict assumptions and are capable of capturing non-linear interactions, making them highly suitable for modeling dynamic land use systems.

Among various machine learning approaches, Artificial Neural Networks (ANN) are one of the most widely used models for predicting land use changes. ANN models are inspired by the structure and functioning of the human brain and consist of interconnected processing units known as neurons. These neurons are organized into layers, including input, hidden, and output layers. The model learns from historical data through a training process, where weights are adjusted iteratively to minimize prediction error. As a result, ANN models are capable of identifying complex, non-linear relationships between different land use classes and their driving factors (Pijanowski et al., 2002).

ANN-based models have been successfully applied in numerous LULC studies and have demonstrated superior performance compared to traditional statistical and rule-based models. Their ability to generalize from training data and adapt to complex spatial patterns makes them particularly effective for predicting urban expansion and land transformation processes.

In addition to ANN, several other machine learning techniques have been widely applied in LULC classification and prediction. Random Forest (RF) is an ensemble learning method that constructs multiple decision trees and combines their outputs to improve prediction accuracy and reduce overfitting. Support Vector Machines (SVM) are powerful classification algorithms that identify optimal decision boundaries between different land use classes, making them highly effective for high-dimensional datasets. Decision Trees provide a simple yet effective approach for classification by splitting data based on decision rules derived from input features.

These machine learning methods have shown improved performance in both classification accuracy and prediction reliability compared to conventional approaches. Their ability to integrate multiple datasets and handle complex spatial interactions has made them increasingly popular in geospatial analysis and environmental modeling.

Overall, machine learning techniques provide a robust and flexible framework for LULC

prediction. However, with the increasing availability of high-resolution spatial data and the growing complexity of urban systems, there is a need for more advanced approaches such as deep learning models, which offer even greater capability in capturing intricate spatial and temporal patterns.

2.6 Deep Learning Techniques for LULC Prediction

Deep learning, an advanced subset of machine learning, has significantly enhanced the capability of land use/land cover (LULC) prediction models by enabling the analysis of complex, high-dimensional spatial data. Unlike conventional machine learning techniques, deep learning models utilize multiple hidden layers to automatically learn hierarchical representations of data, making them highly effective for capturing intricate spatial and temporal patterns (Goodfellow et al., 2016).

Among various deep learning approaches, the Multilayer Perceptron (MLP) is one of the most widely used models for LULC prediction. MLP is a type of feedforward artificial neural network that consists of an input layer, one or more hidden layers, and an output layer. Each neuron in a layer is connected to neurons in the subsequent layer through weighted connections, and the model learns by adjusting these weights during the training process. This structure allows MLP models to effectively model complex and non-linear relationships between different land use classes and their influencing factors.

Deep learning models are capable of processing large volumes of spatial and temporal data, including satellite imagery, geographic variables, and environmental parameters. Their ability to automatically extract meaningful features from raw data eliminates the need for manual feature engineering, thereby improving efficiency and accuracy. As a result, deep learning techniques have become highly effective tools for predicting future land use scenarios and simulating urban growth patterns.

According to LeCun et al. (2015), deep learning has revolutionized pattern recognition and spatial analysis by providing significantly higher accuracy and efficiency compared to traditional and conventional machine learning approaches. These models are particularly useful in handling complex datasets where relationships between variables are highly non-linear and difficult to model using traditional techniques.

The integration of deep learning with Geographic Information Systems (GIS) further

enhances the capability of LULC prediction. GIS provides spatial data management, visualization, and analysis capabilities, while deep learning models perform pattern recognition and prediction. This combined framework enables the development of advanced predictive models that can simulate land transformation processes, identify future urban expansion zones, and support sustainable urban planning.

Overall, deep learning techniques, particularly MLP models, offer a powerful and reliable approach for LULC prediction. Their ability to handle large datasets, capture complex relationships, and generate accurate predictions makes them highly suitable for analyzing dynamic and rapidly changing urban environments.

2.7 LULC Studies in Indian Context

In India, a considerable number of studies have been conducted to analyze land use/land cover (LULC) changes in response to rapid urbanization, population growth, and economic development. Major metropolitan cities such as Delhi, Mumbai, and Bangalore have experienced significant urban expansion over the past few decades, resulting in the large-scale conversion of agricultural land, vegetation, and open spaces into built-up areas (Jat et al., 2008). These transformations have raised serious concerns regarding environmental sustainability, urban infrastructure, and resource management.

Remote sensing and GIS-based approaches have been widely applied in the Indian context for LULC mapping and change detection. Several studies have utilized multi-temporal satellite data to assess urban growth patterns and quantify land use transitions. The findings consistently indicate a rapid increase in built-up areas accompanied by a decline in agricultural land and green cover, highlighting the impact of unplanned urbanization.

In the Gangetic plains, particularly in eastern Uttar Pradesh, LULC changes have been closely linked to hydrological processes and flood dynamics. The region is characterized by flat terrain, dense river networks, and high monsoon rainfall, making it highly susceptible to flooding. Studies conducted in this region have emphasized that alterations in land use patterns, such as the expansion of impervious surfaces and encroachment on natural drainage systems, significantly influence flood behavior and increase vulnerability (Sinha et al., 2014).

Despite the extensive use of remote sensing and GIS in LULC studies across India, most

research efforts are primarily descriptive in nature and focus on historical change analysis. There is a noticeable lack of studies that incorporate advanced predictive modeling techniques, particularly those based on machine learning and deep learning approaches. The application of models such as Artificial Neural Networks (ANN) and Multilayer Perceptron (MLP) remains limited, especially in medium-sized cities like Gorakhpur.

This gap highlights the need for more comprehensive and integrated studies that combine geospatial technologies with advanced computational methods. The adoption of GIS integrated with AI-based approaches can significantly improve the accuracy of LULC prediction and provide valuable insights for sustainable urban planning, environmental management, and disaster risk reduction in the Indian context.

2.8 Research Gap

Based on the comprehensive review of existing literature, several important research gaps have been identified in the field of land use/land cover (LULC) analysis and prediction.

Firstly, although numerous studies have been conducted using remote sensing and GIS for LULC mapping and change detection, the application of advanced deep learning models remains limited, particularly in medium-sized cities such as Gorakhpur. Most existing research is concentrated on large metropolitan areas, leaving smaller yet rapidly growing urban centers relatively underexplored.

Secondly, a majority of previous studies focus primarily on short-term analysis and prediction, typically limited to 10–20 years. There is a noticeable lack of long-term forecasting studies that extend beyond 20–30 years, which are essential for effective urban planning, infrastructure development, and environmental sustainability assessment.

Another significant gap is the insufficient integration of Geographic Information Systems (GIS) with advanced artificial intelligence (AI) and deep learning techniques. While GIS provides strong capabilities for spatial analysis and visualization, its combined use with models such as Artificial Neural Networks (ANN) and Multilayer Perceptron (MLP) is still not fully explored in many studies. This limits the ability to capture complex spatial relationships and reduces the accuracy of predictive modeling.

Furthermore, limited attention has been given to flood-prone regions where LULC changes have a direct and significant impact on disaster risk. In areas such as Gorakhpur, located in

the floodplains of the Rapti River, land use transformations such as the expansion of built-up areas and reduction of natural drainage systems play a crucial role in influencing flood vulnerability. However, existing studies rarely integrate LULC prediction with disaster risk considerations.

In addition, many studies do not adequately incorporate spatial heterogeneity and non-linear interactions among different land use classes, which are critical for accurately modeling urban growth patterns. Traditional models often fail to represent these complexities, highlighting the need for more advanced approaches.

The present study addresses these research gaps by integrating remote sensing, GIS, and deep learning techniques to analyze historical LULC changes and predict future land use scenarios for Gorakhpur city. It emphasizes long-term forecasting, improved model accuracy, and the consideration of environmental implications, particularly in a flood-prone urban setting.

2.9 Summary

The review of existing literature indicates that significant advancements have been made in the field of land use/land cover (LULC) mapping and change detection through the application of remote sensing and Geographic Information Systems (GIS). These techniques have proven to be highly effective in analyzing spatial and temporal land use dynamics, enabling researchers to monitor urban expansion, environmental changes, and landscape transformations with considerable accuracy.

However, traditional prediction models such as Markov Chain and Cellular Automata exhibit limitations in capturing complex spatial interactions and non-linear relationships inherent in dynamic land use systems. As urban environments continue to evolve under the influence of multiple interacting factors, these conventional approaches often fall short in providing highly accurate and reliable future predictions.

The emergence of machine learning and deep learning techniques, particularly Artificial Neural Networks (ANN) and Multilayer Perceptron (MLP), has significantly improved the capability of LULC prediction models. These advanced approaches are capable of handling large datasets, modeling non-linear relationships, and capturing complex spatial patterns, thereby offering enhanced prediction accuracy and reliability.

Despite these technological advancements, a notable research gap persists in the application

of such techniques to medium-sized and environmentally sensitive cities like Gorakhpur. Most existing studies are concentrated on large urban centers and are limited to short-term analysis, with relatively few studies focusing on long-term forecasting and flood-prone regions.

In this context, the present study aims to bridge these gaps by integrating remote sensing, GIS, and deep learning approaches to analyze historical LULC changes and predict future land use scenarios in Gorakhpur city. By emphasizing long-term forecasting and incorporating advanced modeling techniques, the study contributes to improved understanding of urban growth patterns and supports sustainable urban planning and environmental management.

CHAPTER 3

DATA AND METHODOLOGY

3.1 Overview of Data and Methodology

This chapter explains the study area, the datasets used, and the methodological framework adopted to analyse land use/land cover (LULC) transformation and forecasting in Gorakhpur City. The overall workflow was designed to produce comparable LULC maps for multiple years, evaluate their accuracy, quantify change, and then forecast future land-cover dynamics using a deep-learning-based approach. Remote sensing and GIS are widely used for this type of work because satellite archives such as Landsat provide long, consistent records of land-surface conditions, while GIS supports spatial extraction, overlay, change analysis, and model integration (Wulder et al., 2022; USGS, 2022).

3.2 Data Used

The present study used a combination of multi-temporal satellite imagery, ancillary geospatial data, and reference information to analyse land use/land cover (LULC) transformation in Gorakhpur City. The selection of data was based on temporal continuity, spatial suitability, availability, and relevance to urban land-change studies. Since the objective of the research was to identify past and present LULC patterns and to support future forecasting, it was necessary to use datasets that are consistent across time and widely accepted in remote sensing research. For this reason, Landsat imagery was selected as the principal dataset, because the Landsat program provides a long-term, continuous, multispectral record of the Earth's surface and has been extensively used for land-cover mapping and change detection studies .

The main satellite data for the study were obtained from the USGS Earth Explorer platform. Landsat imagery for the selected years, such as Landsat 5 TM for the earlier period and Landsat 8/9 OLI-TIRS for the recent periods, was used to ensure temporal comparability in the analysis. Landsat imagery (30 m resolution) was used for multi-temporal analysis due to its long-term consistency (Wulder et al., 2022). Sentinel-2 data supported validation and training. DEM data from SRTM provided terrain context.

(USGS, 2018). The medium spatial resolution of Landsat is especially appropriate for city-

and district-scale LULC studies because it offers a balance between long historical coverage and acceptable thematic detail. In the case of Gorakhpur City, Landsat data were considered highly appropriate for monitoring the spatial expansion of urban land and the conversion of surrounding agricultural and open land over time.

To support recent interpretation and improve the identification of mixed land-cover features, Sentinel-2 MSI data were also considered as supplementary data. Sentinel-2 is an important source for modern land-cover studies because it provides 13 spectral bands at 10 m, 20 m, and 60 m spatial resolutions, together with a 5-day revisit period under the twin-satellite mission configuration (ESA, 2024). Compared with Landsat, Sentinel-2 offers finer spatial detail for recent years, which is especially useful in urban fringe areas where built-up patches, vegetation, and open land often appear in fragmented form. Phiri et al. (2020) also noted that Sentinel-2 has significantly improved land-cover/use monitoring, particularly in urban areas, crops, forests, and water resources. Therefore, in the present study, Sentinel-2 was not used as the main temporal dataset, but rather as a supporting dataset for recent-year visual interpretation, training sample refinement, and validation of classified outputs.

In addition to satellite imagery, the study used ancillary spatial data to strengthen the GIS and image-processing workflow. Administrative boundary data were required to define the exact extent of Gorakhpur City and to clip the satellite imagery to the study boundary before classification. This step was necessary to ensure that area statistics and thematic maps reflected only the selected study area. A Digital Elevation Model (DEM) was also used as supporting information for terrain interpretation. For this purpose, the Shuttle Radar Topography Mission (SRTM) dataset was considered suitable, because NASA Earthdata states that SRTM collected topographic data over nearly 80% of Earth's land surface, creating one of the first near-global elevation datasets (NASA Earthdata, 2026). In a flood-prone and low-lying region like Gorakhpur, DEM data are useful for understanding terrain conditions, drainage influence, and the environmental background of LULC transformation. Apart from primary and ancillary datasets, reference data were also important for classification and interpretation. In supervised LULC studies, reference information helps in selecting representative training samples and in distinguishing classes that may show similar spectral behaviour. In the present study, reference support from visual interpretation and recent high-resolution imagery was considered necessary for properly identifying the major

classes, namely agricultural land, built-up area, vegetation/forest, wasteland, and water bodies. Thus, the integration of Landsat data, Sentinel-2 support data, DEM, administrative boundaries, and reference information created a reliable database for classification, change detection, and subsequent forecasting analysis. Overall, the selected datasets were scientifically appropriate for the present study because they combined long-term temporal coverage, acceptable spatial resolution, and established use in previous remote sensing research (Wulder et al., 2022; Phiri et al., 2020).

Type of Data	Source	Resolution / Scale	purpose
Landsat 5 TM	USGS Earth Explorer	30 m	preparation of historical LULC map for 2005
Landsat 8 OLI	USGS Earth Explorer	30 m	Preparation of LULC map for 2015 and temporal comparison
Landsat 8/9 OLI	USGS Earth Explorer	30 m	Preparation of recent LULC map for 2025 and future trend assessment
Administrative Boundary	Survey of India / GADM / digitized boundary	1:50,000 or available scale	Defining the study area boundary of Gorakhpur City
High-Resolution Imagery	Google Earth Pro / visual interpretation	30 m	Training sample selection, validation, and ground reference support
Reference Samples	Field knowledge / Google Earth / visual interpretation		Class identification and accuracy assessment
GIS Software	ArcGIS / QGIS		Image clipping, spatial analysis, map preparation, and area calculation
Analytical/ Modelling Tools	Google Earth Engine / Python / TensorFlow / Keras / CA–Markov tool		Image preprocessing, classification, change detection, and forecasting

Table 3.1 Summary of Data Sources and Analytical Inputs Used for Land Use/Land Cover Transformation and Forecasting in Gorakhpur City.

3.3 Data Pre-processing

Data pre-processing was carried out to make the satellite images suitable for land use/land cover classification and multi-temporal comparison. In LULC studies, preprocessing is necessary because images acquired on different dates may contain atmospheric disturbance, cloud contamination, and geometric variation, which can reduce the reliability of classification and change analysis. Earlier research has shown that atmospheric effects and image inconsistency can influence classification and change-detection results if they are not handled properly (Song et al., 2001).

In the present study, suitable Landsat scenes were selected from the USGS archive by considering low cloud cover and similar seasonal conditions. This was done to reduce seasonal differences between years and to improve temporal comparability. Landsat Collection 2 Level-2 products were used because they provide scientifically prepared surface reflectance data for land analysis, which makes them appropriate for land-cover studies (USGS, 2022).

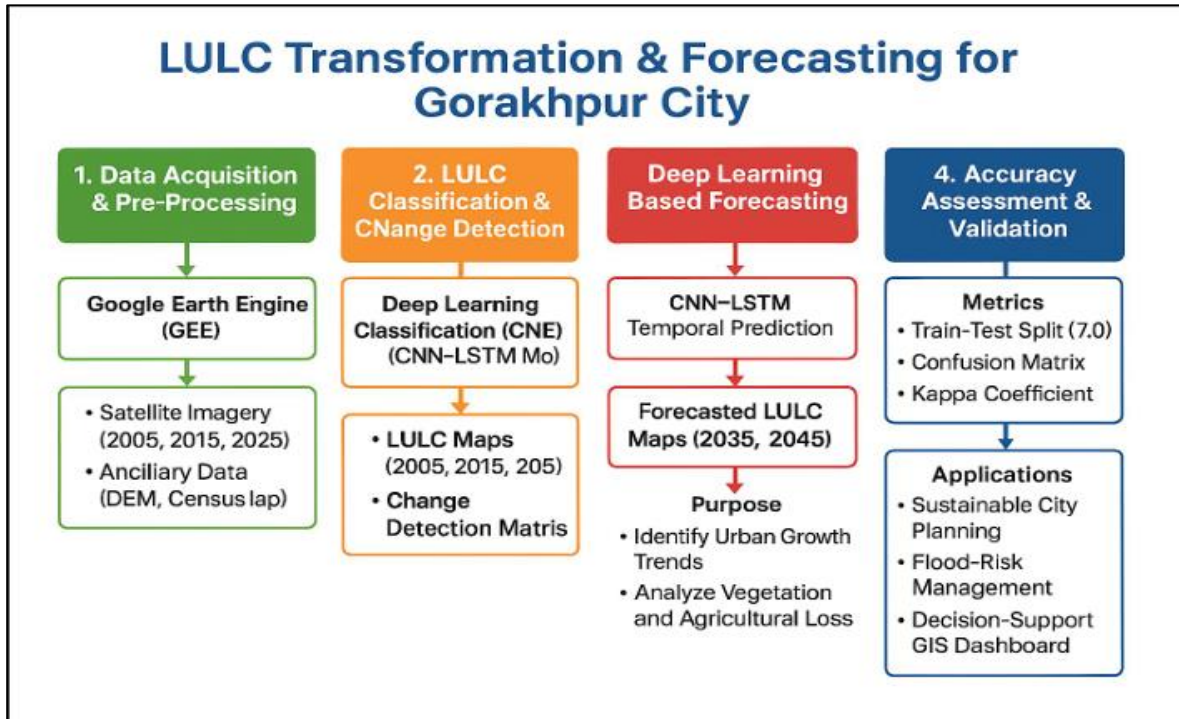


Figure 3.1 Overall methodological framework adopted for land use/land cover transformation analysis and forecasting in Gorakhpur City using GIS, remote sensing, deep learning classification, and CA–Markov simulation.

After image selection, atmospheric and radiometric correction were addressed through the use of surface reflectance products. This step was important because it improved the comparability of spectral values between dates and helped in distinguishing major classes such as built-up land, agricultural land, vegetation, wasteland, and water bodies. Where Sentinel-2 data were used as supporting imagery, Level-2A products were preferred because they provide atmospherically corrected surface reflectance derived from Level-1C data (Copernicus, 2026).

Cloud masking and quality screening were also performed before classification so that pixels affected by cloud, shadow, or poor image quality did not influence the final output. After that, the required spectral bands were prepared and the imagery was clipped to the Gorakhpur City boundary for focused analysis. This step improved processing efficiency and ensured that the statistics represented only the actual study area. Since Landsat has been widely accepted as a reliable long-term dataset for land monitoring, such preprocessing helps in producing more consistent and dependable LULC maps (Wulder et al., 2022).

3.4 Land Use/Land Cover Classification

LULC classification was undertaken to convert the pre-processed imagery into a set of thematic classes that represent the major surface characteristics of Gorakhpur City. For this study, a simplified but locally meaningful classification structure was adopted, consisting of built-up area, agricultural land, vegetation/forest, wasteland/barren land, and water bodies. A simplified scheme is suitable for medium-resolution satellite mapping because it reduces class ambiguity while still capturing the dominant landscape transitions relevant to urban expansion and environmental planning. The long-standing USGS/Anderson land-use and land-cover classification framework and later Landsat classification reviews both support the idea that Level I/II style thematic groupings remain practical for satellite-based mapping (Anderson et al., 1976; Phiri and Morgenroth, 2017).

A **supervised classification** approach was adopted. In this method, the analyst identifies representative training areas for each class, and the classifier then assigns image pixels to the class they most closely match. Landsat classification reviews show that supervised methods have been used extensively through the history of Landsat-based mapping, and that classifier choice, training quality, and input preparation all influence final map accuracy (Phiri and

Morgenroth, 2017; Khatami et al., 2016).

For the present study, Maximum Likelihood Classification (MLC) is an acceptable methodological choice because it remains one of the most established supervised classifiers in remote sensing. Review literature notes that MLC has been widely applied in LULC studies, and technical descriptions of the method explain that it assigns each pixel to the class with the highest probability on the basis of training-data statistics. In examiner terms, this means that MLC is suitable when the objective is to produce interpretable, comparable class maps from multispectral imagery rather than to test a large suite of machine-learning algorithms (Chughtai et al., 2021; Hagner and Reese, 2007)

Class	Description
Built-up area	Residential, commercial, roads, and other constructed surfaces
Agricultural land	Cropland and related agricultural fields
Vegetation/Forest	Tree cover, plantation, and dense vegetation patches
Wasteland/Barren land	Open land, fallow areas, and sparsely vegetated surfaces
Water bodies	Rivers, ponds, lakes, and other surface water features
Built-up area	Residential, commercial, roads, and other constructed surfaces

3.5 Change Detection Analysis

After classification and validation, the study used **post-classification comparison** to identify and quantify LULC change between the selected years. Change-detection reviews describe post-classification comparison as one of the most widely used methods for LULC studies because it enables direct **from-to class comparison** between independently classified maps. It is especially useful when the aim is to calculate gain, loss, conversion, and net change for each land-cover category (Chughtai et al., 2021; El-Hattab et al., 2016).

In this study, the change-detection process involved comparing the classified maps for **2005–2015**, **2015–2025**, and **2005–2025**. The analysis should report class-wise area statistics, percentage change, net change, and, where appropriate, annual rate of change. A **change matrix** should also be prepared to show how much area moved from one class to another, for example from agricultural land to built-up area or from vegetation to barren land. This is methodologically important because simple area totals do not show the direction of conversion, whereas the change matrix does (Chughtai et al., 2021).

Where the report discusses method selection, you can add that post-classification methods are often preferred because they can produce more explicit thematic change information than

direct spectral differencing, although their reliability still depends on the accuracy of the individual classified maps. Comparative studies have shown that post-classification approaches can perform strongly in identifying changed and unchanged areas when well-classified source maps are available (Afify, 2011; El-Hattab et al., 2016).

3.6 Forecasting Using Deep Learning

To extend the analysis beyond historical change detection, the study incorporated a deep-learning-based forecasting stage to predict future LULC conditions. This part of the methodology is justified because recent remote-sensing research has increasingly used deep learning for land-cover classification and spatiotemporal prediction, especially where change processes are nonlinear and influenced by multiple variables. Reviews of deep learning in LULC mapping show clear growth in the use of neural-network-based methods during the last decade (Vali et al., 2020; Li et al., 2024).

A practical and academically defensible forecasting design for this project is an LSTM-based or MLP-LSTM-based framework. Recent land-use prediction studies explain that LSTM models use memory cells and gate mechanisms to retain sequence information and are therefore well suited to modelling complex temporal dynamics. Wang et al. (2023) reported that an MLP-LSTM framework retained spatiotemporal information effectively and outperformed comparison models in their case study, while Alkaraki et al. (2024) integrated land-cover maps with NDVI, NDBI, and topographic variables for future prediction.

For this study, the forecasting workflow can be written as follows: first, the classified LULC maps for the selected years were converted into model-ready raster layers; second, supplementary predictor layers such as vegetation condition, built-up intensity, and topographic information were organized; third, the model was trained using the temporal sequence of observed land-cover states; fourth, the trained model generated a future LULC surface for the target year; and finally, the model was validated through **back-prediction**, such as using earlier maps to predict a known later map and comparing the prediction against the actual classified map. This type of validation design is consistent with recent land-use prediction studies that assess model performance before producing future scenarios (Wang et al., 2023; Alkaraki et al., 2024).

For, keep the forecasting subsection very explicit. State the input years, the predictor

variables, the model architecture, the training-validation logic, and the target forecast year. Avoid vague wording such as “deep learning was applied for future prediction.” Instead write: “An LSTM-based spatiotemporal forecasting model was used to learn temporal transitions among classified LULC states and selected ancillary predictors, and the model output was used to generate the predicted LULC map for [target year].” That wording is far stronger scientifically.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Overview of Result and Discussion

This chapter presents the results of land use/land cover (LULC) classification, temporal change analysis, and future prediction for Gorakhpur City for the years 2005, 2015, 2025, and the forecast years 2035, 2045, and 2055. The chapter is structured to explain the spatial distribution of the major LULC classes, quantify the area change recorded in the supplied statistics, interpret the direction of change from the maps, and discuss the likely environmental and planning implications. This structure is important because Gorakhpur is not only an expanding urban centre but also a city located in the basin of the Rapti and Rohini rivers with a bowl-shaped geographical setting, which makes land transformation especially significant for drainage, water sensitivity, and urban planning (District Gorakhpur).

The results of the present study show a clear and progressive transformation from an agriculture-dominated landscape toward a more urbanized built-up system. The strongest pattern visible in both the maps and the statistics is the persistent increase in built-up area and the corresponding decline in agricultural land. This broad pattern agrees with published studies on Gorakhpur urban area, which also report rapid built-up growth, reduction of agricultural land, and rising landscape pressure associated with urban expansion. (Kumar, Naresh, 2024).

4.2 Land Use/Land Cover Status of Gorakhpur City

4.2.1 LULC status in 2005

The 2005 map represents the baseline land-cover condition of the study area. At this stage, agricultural land was the dominant class with an area of 145.6 km², followed by built-up area with 42.8 km², wasteland with 22.5 km², forest with 18.4 km², and water bodies with 11.7 km². In percentage terms, agricultural land accounted for approximately 60.4% of the mapped area, while built-up land contributed about 17.8%. This shows that the study area in 2005 was still primarily structured around an agrarian land system with a relatively compact urban core.

The spatial arrangement of the classes in Figure 5.1 indicates that built-up land was concentrated mainly in the central part of Gorakhpur City, with limited outward expansion. Agricultural land formed a broad continuous matrix around the urban core, especially across the northern, eastern, and western parts of the study area. Forest was present in relatively distinct patches rather than as a continuous cover, while water bodies were confined to a few stable features, including the prominent lake zone near the central-southern part of the city. Wasteland occurred as scattered transitional patches, particularly around the urban fringe.

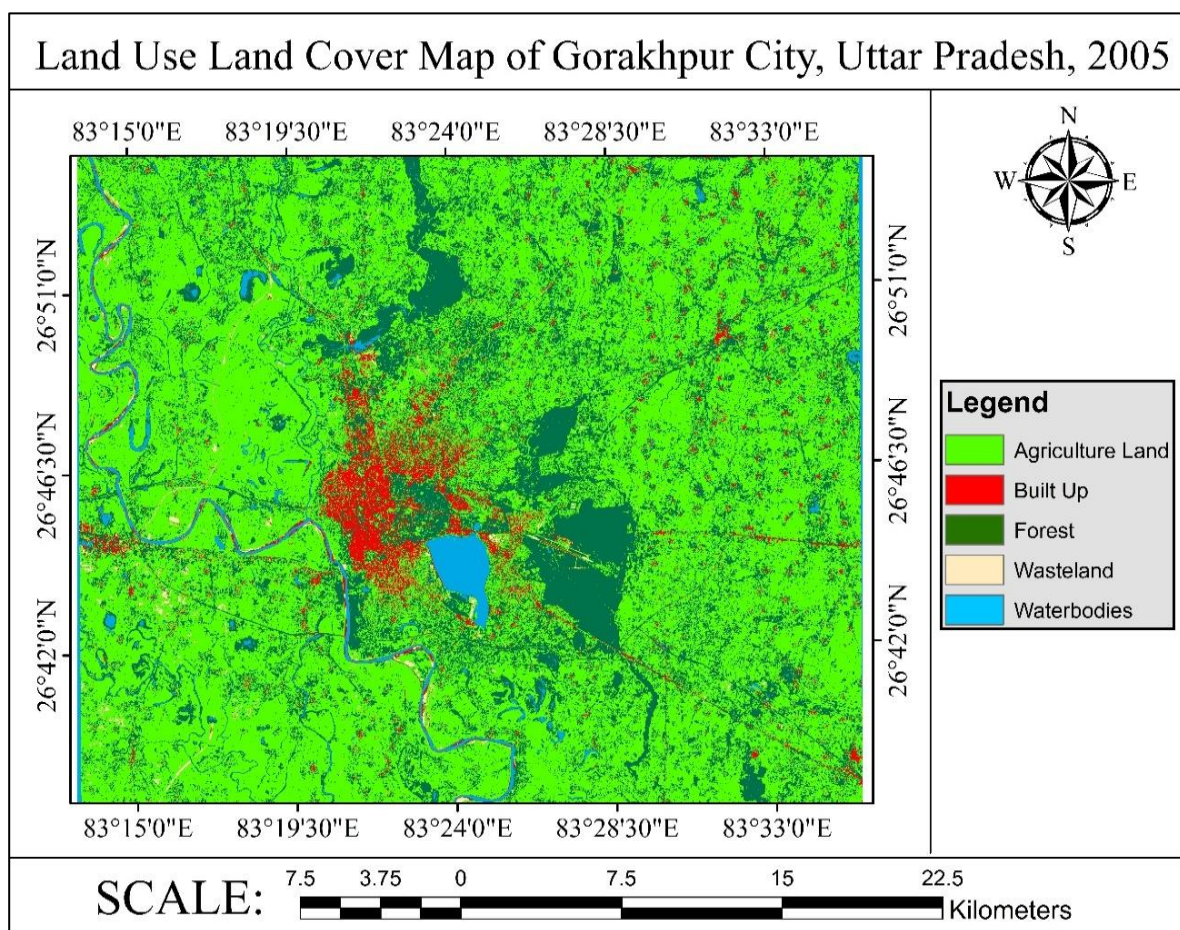


Figure 4.1 Land Use/Land Cover map of Gorakhpur City, Uttar Pradesh, for the year 2005 derived from Landsat imagery using supervised Maximum Likelihood Classification, showing the baseline spatial distribution of built-up area, agricultural land, forest, wasteland, and water bodies

LULC Class	Area (km ²)
Agricultural Land	145.6
Built-up Area	42.8
Forest	18.4
Wasteland	22.5
Water Bodies	11.7

Table 4.1 Area statistics of LULC classes in Gorakhpur City for 2005

From an urban morphology perspective, the 2005 map suggests a compact-city pattern, in which the main built-up concentration remained spatially bounded and did not yet dominate the peri-urban landscape. This interpretation is important because it establishes the baseline against which all later transformation is measured. In Gorakhpur, such a baseline matters especially because later urban growth has been linked in the literature to increasing imperviousness and broader landscape modification (hoo, Santanu & Tyagi, Nutan. (2022)).

4.2.2 LULC status in 2015

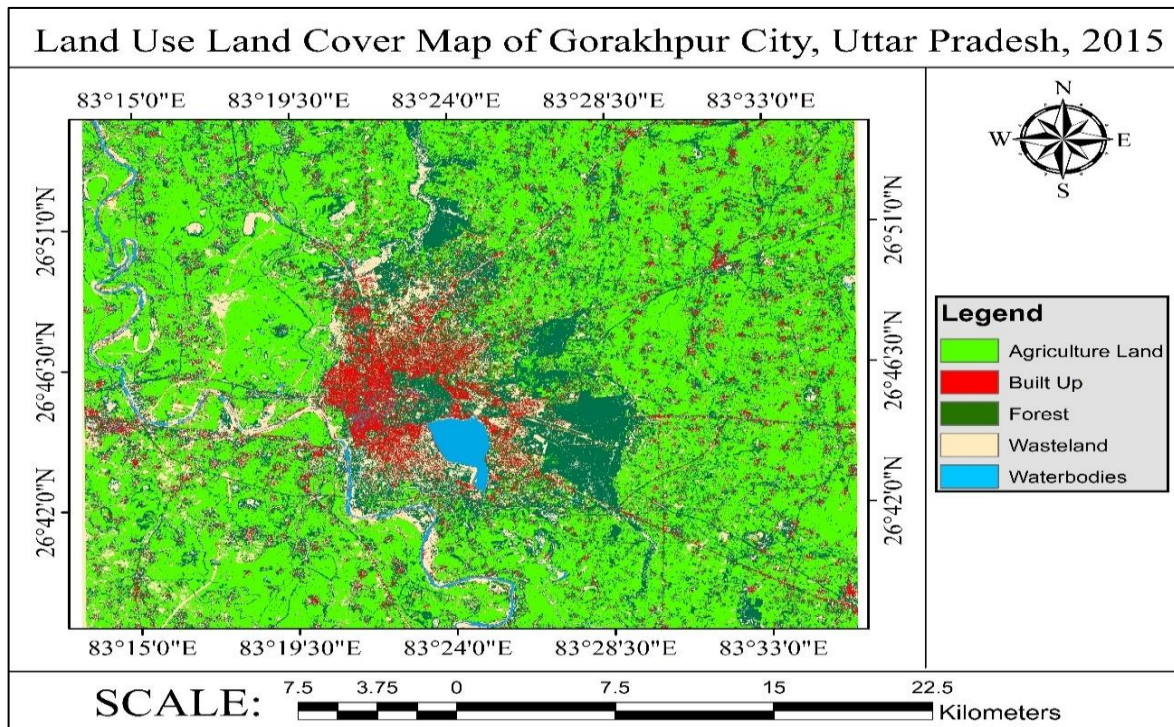


Figure 4.2 Land Use/Land Cover map of Gorakhpur City, Uttar Pradesh, for the year 2015 showing transitional urban expansion and associated land-cover changes.

LULC Class	Area (km²)
Agricultural Land	132.4
Built-up Area	56.7
Forest	19.2
Wasteland	20.1
Water Bodies	12.4

Table 4.2 Area statistics of LULC classes in Gorakhpur City for 2015

The 2015 map shows a clear transitional stage in the land transformation of Gorakhpur City. By 2015, built-up area increased to 56.7 km², while agricultural land declined to 132.4 km². Forest showed a slight rise to 19.2 km², wasteland decreased to 20.1 km², and water bodies increased marginally to 12.4 km². Compared with 2005, the most notable numerical change was the increase of 13.9 km² in built-up land and the decrease of 13.2 km² in agricultural land.

The map indicates that the built-up zone was no longer limited to a compact core. Instead, it expanded outward and became more dispersed, especially along the road-linked fringe areas. This suggests that the city had entered a stronger phase of peri-urban growth, in which the central core continued to expand while adjacent non-urban land began to fragment. Agricultural land still remained the largest class, but its continuity weakened near the urban margin. This pattern is visible in the increased red patches spreading into the surrounding green agricultural surface.

A second important observation is that the 2015 map shows greater heterogeneity in the urban fringe than the 2005 map. This means that the city was not only expanding outward but also generating a more mixed peri-urban landscape made up of built-up patches, remnant agricultural parcels, wasteland, and fragmented vegetation. Such a pattern is commonly associated with early-to-middle stages of urban sprawl. For Gorakhpur specifically, published work has noted that rapid urbanization has been putting pressure on natural resources and consuming agricultural land at the city's periphery, which is consistent with the transition visible in your 2015 map (Sahoo, Santanu & Tyagi, Nutan. 2022).

4.2.3 LULC status in 2025

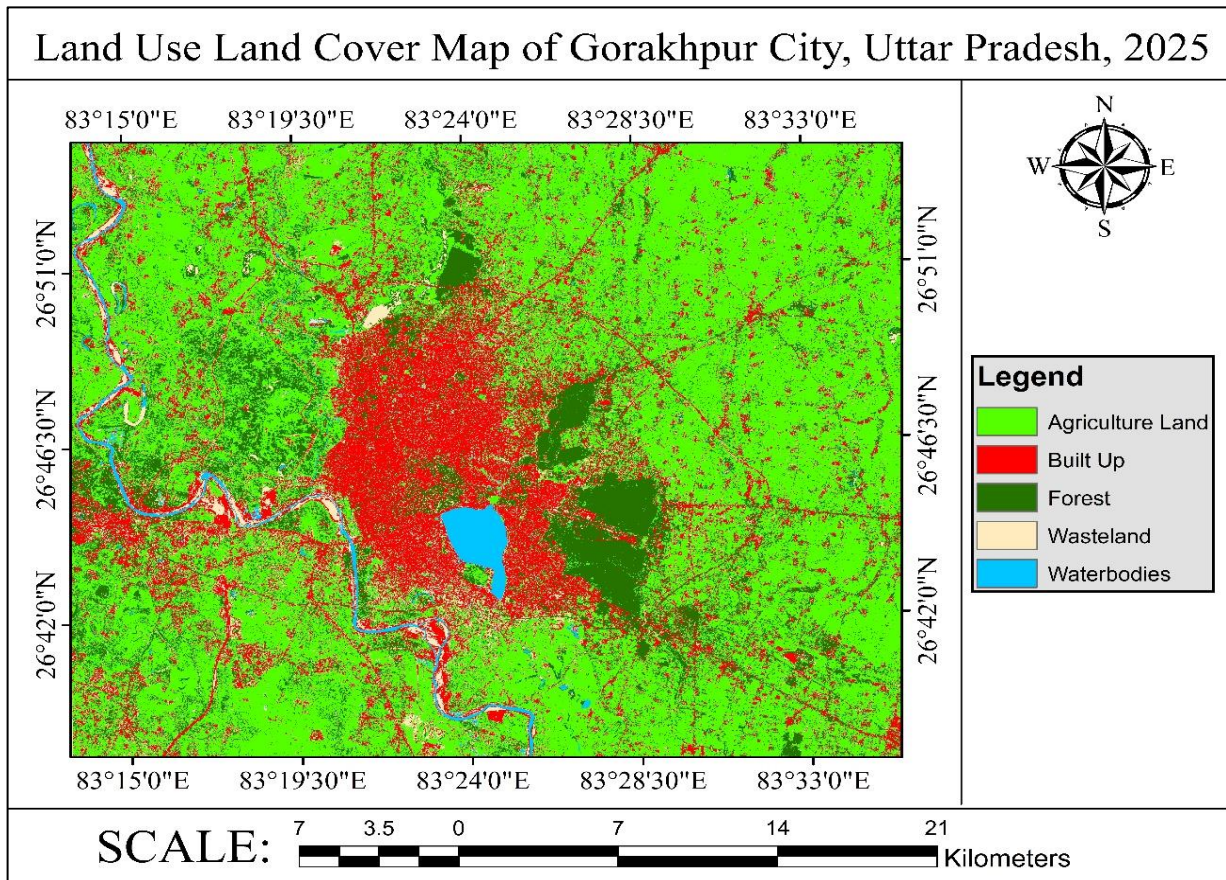


Figure 4.3. Land Use/Land Cover map of Gorakhpur City, Uttar Pradesh, for the year 2025 derived from Landsat 8/9 OLI imagery using supervised Maximum Likelihood Classification, showing significant urban expansion and increasing land transformation.

LULC Class	Area (km ²)
Agricultural Land	118.3
Built-up Area	73.5
Forest	17.9
Wasteland	19.6
Water Bodies	11.2

Table 4.3 Area statistics of LULC classes in Gorakhpur City for 2025

The 2025 map represents the most urbanized condition among the observed years. By this time, built-up area increased to 73.5 km², while agricultural land declined further to 118.3 km². Forest decreased slightly to 17.9 km², wasteland fell to 19.6 km², and water bodies

declined to 11.2 km². Built-up land thus gained 30.7 km² between 2005 and 2025, whereas agricultural land lost 27.3 km² over the same period.

The visual pattern of Figure 5.3 shows that urban growth became both more extensive and more consolidated by 2025. The central city appears denser, and the red built-up surface extends more continuously into the surrounding fringe. In comparison with 2015, the built-up system in 2025 is less fragmented and more spatially dominant. Agricultural land, although still the largest class, has clearly lost continuity around the city, especially near the main urban mass and along growth corridors. The forest patches remain visible, but they are increasingly enclosed by the expanding built-up matrix.

This stage of transformation is significant because it reflects not just outward spread but also urban densification and infill. The change is therefore not limited to edge expansion alone; it also includes the strengthening of the urban interior. Published research on Gorakhpur urban area similarly reports that built-up area showed the highest positive change while agricultural land recorded the highest decline, and that this transformation contributed to increasing impervious surfaces and higher land surface temperature (Sahoo, Santanu & Tyagi, Nutan. 2022).

4.3 Comparative Area Statistics and Trend Analysis

The comparative statistics reveal that the dominant land transformation in the study area was the conversion of a primarily agriculture-based landscape into a more urbanized system. Over the full 2005–2025 period, built-up area increased from 42.8 km² to 73.5 km², recording a net gain of 30.7 km² or 71.73%. In contrast, agricultural land declined from 145.6 km² to 118.3 km², giving a net loss of 27.3 km² or 18.75%. Forest recorded only a small net decline of 0.5 km², wasteland decreased by 2.9 km², and water bodies decreased by 0.5 km².

These figures indicate that the most active land competition in Gorakhpur occurred mainly between built-up land and agricultural land. The other classes changed more modestly in total area, although their spatial position and fragmentation may still have changed substantially. In practical terms, this means that urban expansion was the strongest and most systematic process, while the broader ecological structure became increasingly embedded within an urbanizing matrix.

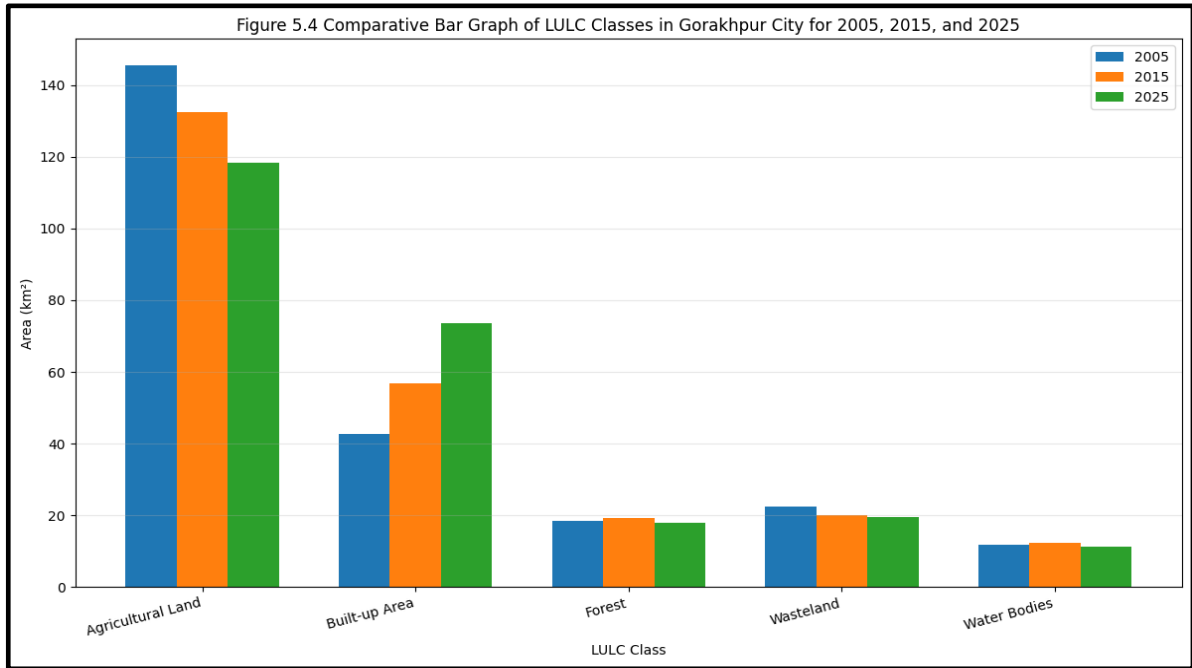


Figure 4.4 The comparative bar graph clearly shows that agricultural land declined steadily from 2005 to 2025, while built-up area increased continuously during the same period, indicating rapid urban expansion in Gorakhpur City. Forest, wasteland, and water bodies showed relatively minor fluctuations compared to the dominant agricultural-to-built-up transformation.

A second important result is that urban growth appears to have accelerated in the later decade. Between 2005 and 2015, built-up land increased by 13.9 km². Between 2015 and 2025, it increased by 16.8 km². Agricultural loss also became slightly stronger in the second decade, decreasing by 14.1 km² from 2015 to 2025, compared with 13.2 km² from 2005 to 2015. This indicates that the pace of urban encroachment into the peri-urban landscape intensified after 2015.

The trend observed in the present dataset agrees with earlier studies from Gorakhpur, which likewise found that built-up land was the fastest-growing class and agricultural land the main declining class under rapid urbanization (Sahoo, Santanu & Tyagi, Nutan. 2022).

4.4 Spatial Change Detection and Landscape Interpretation

4.4.1 Change between 2005 and 2015

The first decade shows the transition from a compact urban form toward a more spatially dispersed growth pattern. The main change visible in the map sequence is the outward spread of built-up land from the central core into adjacent agricultural areas. Although the class statistics alone do not provide a formal class-to-class transition matrix, the near-simultaneous increase in built-up land and decrease in agricultural land strongly suggests that agricultural land was a major source of urban expansion during this period.

The urban fringe in 2015 also became more fragmented than in 2005, which indicates an early sprawl phase. Small and medium built-up patches appear around the main city, creating a patchier spatial structure. This kind of fragmented expansion is typically associated with corridor development, peri-urban settlement growth, and infrastructure-led land conversion.

4.4.2 Change between 2015 and 2025

The second decade shows a stronger and more spatially integrated urban expansion than the first. Built-up land increased more rapidly, and the map suggests a combination of edge expansion, infill, and corridor strengthening. The central built-up mass became denser, and many peripheral patches appear to have merged into a more continuous urban form. Agricultural land continued to shrink, particularly around the urban margin.

This indicates that urban transformation in Gorakhpur moved from a transition stage to a stronger consolidation stage after 2015. In other words, the city did not merely continue to sprawl; it also intensified internally. This is important because densification and continuous impervious development can alter runoff behaviour, thermal conditions, and open-space availability more strongly than fragmented low-density growth alone.

4.4.3 Overall change between 2005 and 2025

Taken together, the 2005, 2015, and 2025 maps show a core-expansion and peri-urban conversion model of landscape transformation. In 2005 the built-up area was compact. In 2015 it spread outward and fragmented the fringe. In 2025 it became both larger and more consolidated. The dominant spatial shift was therefore from a rural-urban contrast toward a more extended urban field.

This long-term pattern is significant for Gorakhpur because the city's physical setting makes it sensitive to land-use pressure. Official district information identifies Gorakhpur as a bowl-

shaped city located in the basin of the Rapti and Rohini rivers . In such a setting, continuous expansion of impervious surfaces can have stronger implications for drainage stress, flood sensitivity, and land-use conflict than in less water-sensitive urban environments.

LULC Class	2005 Area (km ²)	2015 Area (km ²)	2025 Area (km ²)	Net Change (2005–2025) (km ²)	% Change
Agricultural Land	145.6	132.4	118.3	-27.3	-18.75%
Built-up Area	42.8	56.7	73.5	+30.7	+71.73%
Forest	18.4	19.2	17.9	-0.5	-2.71%
Wasteland	22.5	20.1	19.6	-2.9	-12.89%
Water Bodies	11.7	12.4	11.2	-0.5	-4.27%

Table 4.4. Area statistics of major Land Use/Land Cover (LULC) classes in Gorakhpur City for the years 2005, 2015, and 2025, including net change (2005–2025) and percentage change, highlighting significant built-up expansion and agricultural land decline.

4.5 Future LULC Prediction

The forecasted land use/land cover (LULC) results for Gorakhpur City were interpreted from the predicted maps prepared for the years 2035, 2045, and 2055 using Artificial Neural Network / Multilayer Perceptron (ANN/MLP) and Multilayer Perceptron (MLP) approaches. In LULC research, predictive modelling is widely used to understand the likely future direction of urban expansion and landscape conversion, and review studies note that LULC modelling is an important tool for examining future urban growth scenarios and their spatial implications (Gaur et al., 2023).

A broad visual reading of all three forecast years shows a consistent pattern: built-up area is expected to remain the dominant expanding class around the existing urban core, while agricultural land continues as the major outer matrix but becomes increasingly interrupted by urban patches. Forest remains concentrated in a few localized blocks, and water bodies persist mainly in their established positions. This means that the prediction maps do not suggest a reversal of the historical trend observed from 2005 to 2025; instead, they indicate a continuation of the same urbanization trajectory into future decades. This interpretation is

also methodologically reasonable because neural-network-based LULC prediction is commonly used to simulate probable future land conversion, especially where historical urban growth has already shown a strong directional trend (Al-Dousari et al., 2023; Gaur et al., 2023).

4.5.1 Forecasted LULC map for 2035

The 2035 forecast map shows that the central built-up zone is predicted to expand further and remain spatially dominant in the middle of the study area. In both the ANN/MLP and MLP panels, the red built-up class spreads outward from the already urbanized core and occupies a larger continuous area than what was visible in the 2025 observed map. This indicates that the city is likely to continue growing through both edge expansion and internal consolidation. The agricultural class still covers a large part of the surrounding region, but its continuity is visibly broken by numerous built-up patches, particularly around the central and corridor-linked fringe. The 2035 map also suggests that the future growth of Gorakhpur may not be restricted only to the municipal centre. Rather, the expansion appears to diffuse into the surrounding peri-urban belt, producing a landscape in which agricultural land remains present but is increasingly fragmented by red urban insertions. This kind of pattern is typical of urban growth forecasts in which past built-up expansion has already established a strong transition path from agricultural to urban land use (Al-Dousari et al., 2023).

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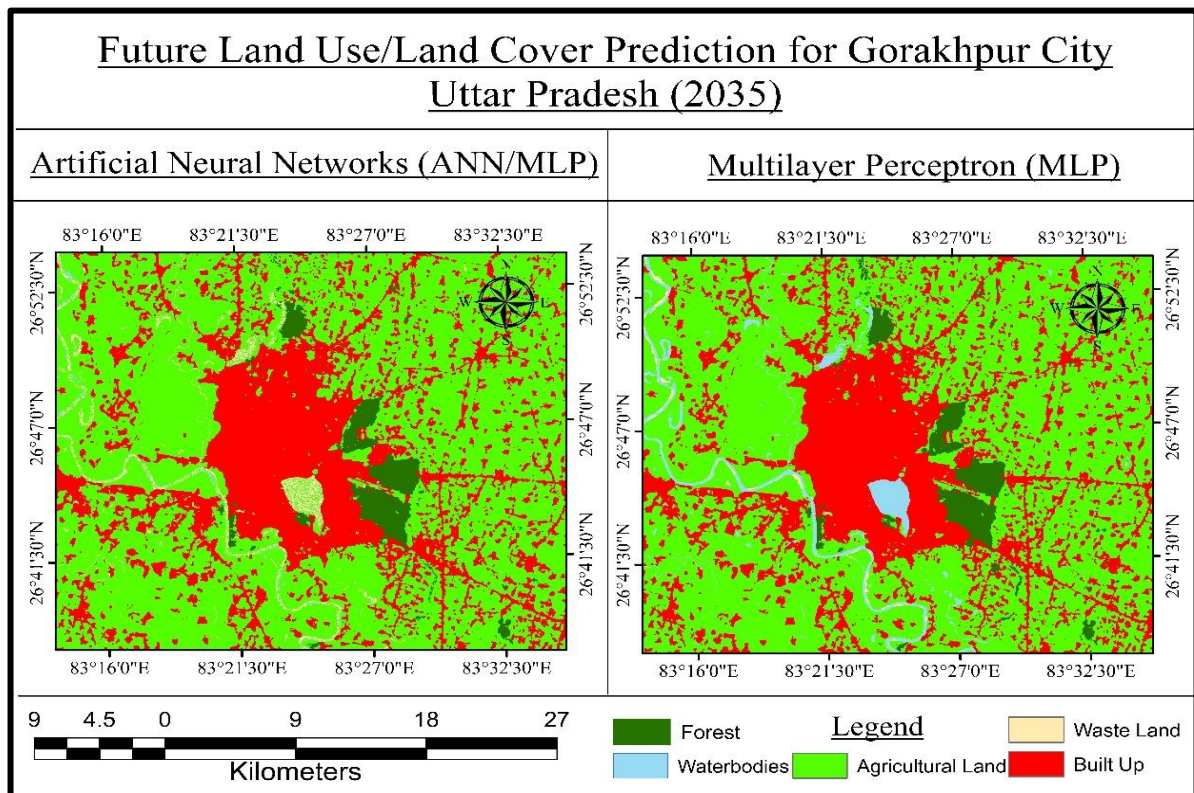


Figure 4.5 Future land use/land cover prediction for Gorakhpur City for 2035 using ANN/MLP and MLP models.

This kind of pattern is typical of urban growth forecasts in which past built-up expansion has already established a strong transition path from agricultural to urban land use (Al-Dousari et al., 2023).

4.5.2 Forecasted LULC pattern for 2045

The 2045 forecast map continues the same structural trend visible in the 2035 prediction. Built-up land remains highly concentrated in the centre and appears to persist as a broad red mass extending toward the surrounding urban fringe. Agricultural land still dominates the outer landscape, but it is again broken by urban patches and corridor-like development. Forest patches remain visible mainly in the eastern and northeastern portions of the map, while water bodies continue to occupy localized low-lying zones and the major lake feature near the city. At the scale of the image you provided, the 2045 forecast is visually very similar to the 2035

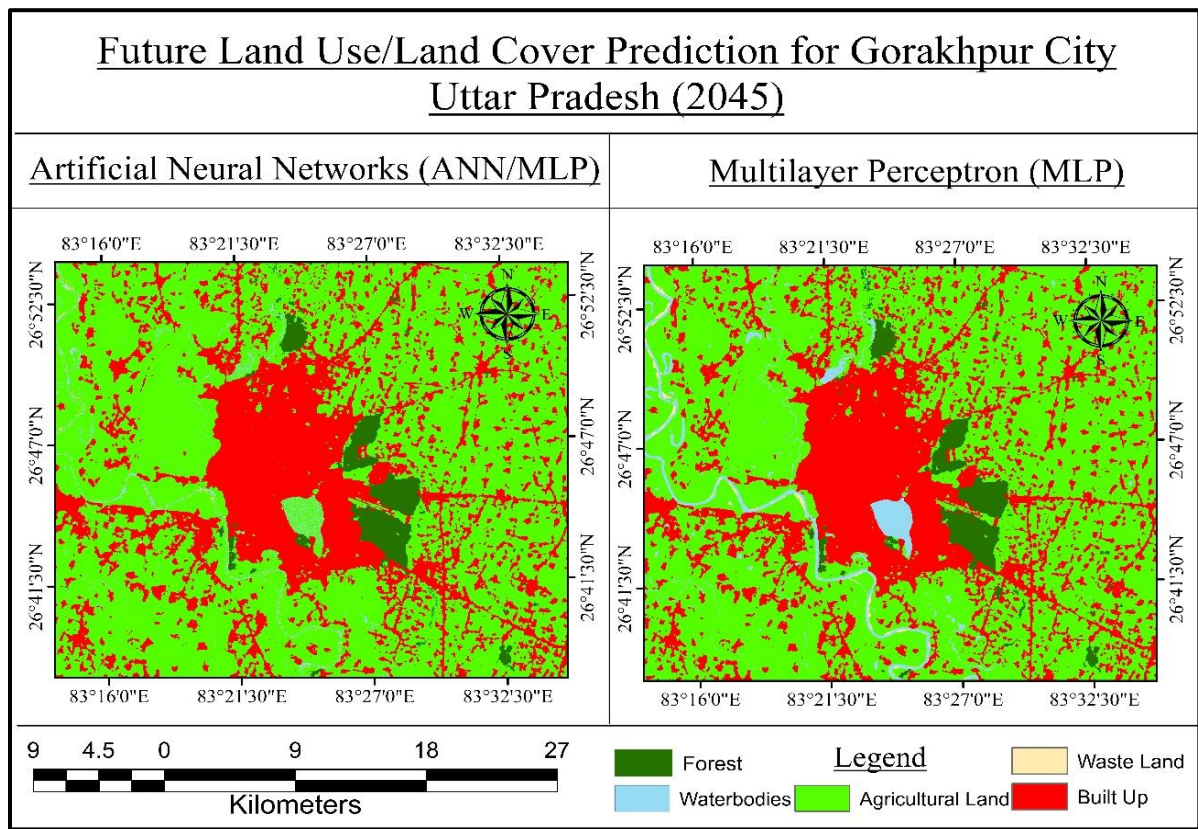


Figure 4.6 Forecasted land use/land cover map of Gorakhpur City for the year 2045 generated using ANN/MLP and MLP models.

forecast. This is an important point and should be stated clearly in the report. It suggests either that the model predicts incremental rather than dramatic additional spatial change between 2035 and 2045, or that the amount of change occurring in that interval is not easily distinguishable at the exported map scale.

In LULC forecasting, this is not unusual, because predictive outputs often show strong early allocation of probable urban-growth zones, while later maps reflect more gradual consolidation of the same trend (Gaur et al., 2023).

Another important interpretation from the 2045 panels is the persistence of green agricultural land around the urban edge, even though it is increasingly perforated by built-up patches. This means that the model is not predicting complete replacement of agriculture by urban land by 2045. Instead, it points toward a mixed peri-urban landscape, where agriculture continues to exist but under stronger pressure from expanding urban use. From a planning

perspective, this is a crucial result because it identifies the peri-urban belt as the primary zone of future land-use conflict.

4.5.3 Forecasted LULC pattern for 2055

The 2055 forecast map indicates the continued dominance of the same long-term pattern: a large and consolidated built-up core surrounded by an agricultural matrix that is increasingly fragmented by urban expansion. The central built-up zone remains strong in both ANN/MLP and MLP outputs, and the outer parts of the study area continue to show numerous red patches distributed through the green agricultural background. This suggests that by 2055 the future land system of Gorakhpur may become even more strongly characterized by **urban-rural intermixing** rather than by a simple compact city surrounded by continuous farmland.

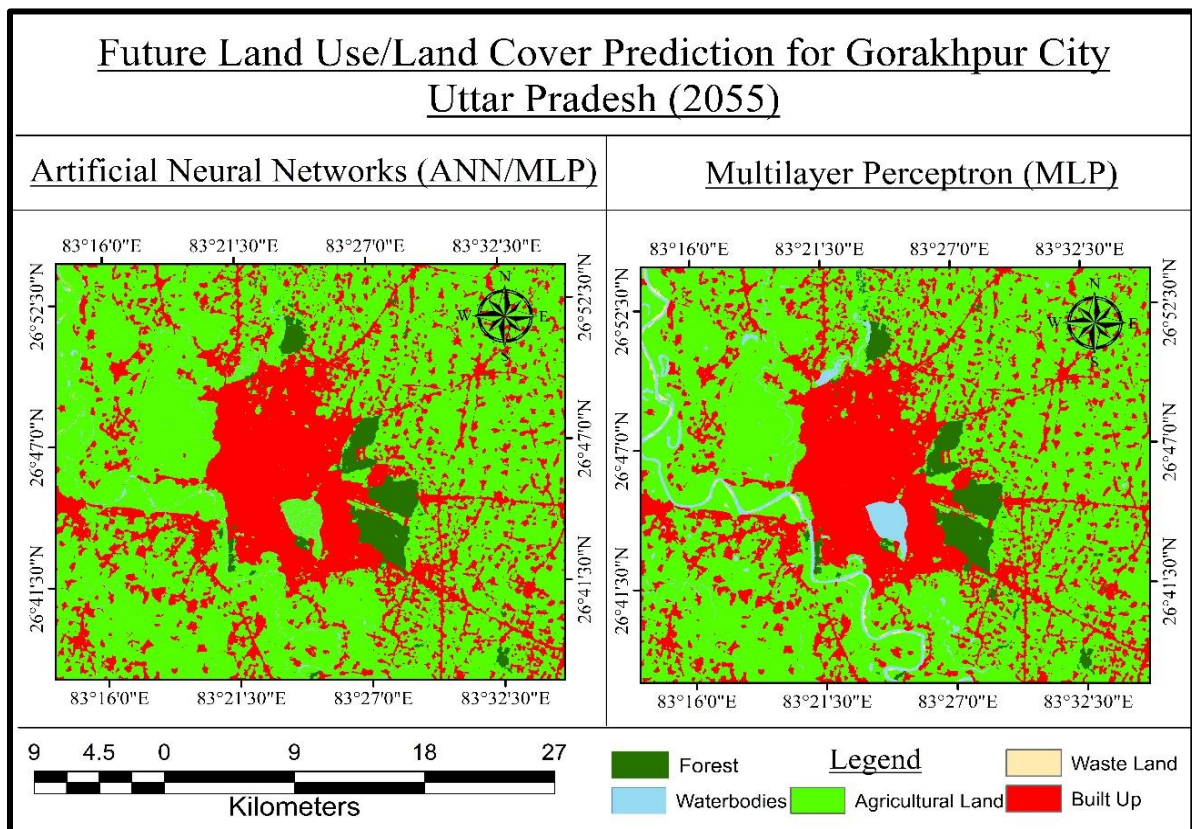


Figure 4.7 Forecasted land use/land cover map of Gorakhpur City for the year 2055 generated using ANN/MLP and MLP models.

As in the 2045 figure, the 2055 map is also visually close to the earlier prediction maps,

especially at the presentation scale used in the figure. This similarity should be interpreted carefully. It does not necessarily mean that no change is predicted after 2035; rather, it may indicate that the models allocate the major future urban growth pattern early and then retain that broad structure through later forecast years. Since no class-wise forecast area table has been provided here, the interpretation of 2055 must remain mainly spatial and qualitative. The 2055 forecast nevertheless strengthens one central conclusion: the future landscape of Gorakhpur is expected to remain under strong urban expansion pressure, with agriculture still present but increasingly disrupted by built-up growth. This agrees with the general purpose of neural-network-based LULC modelling, which is to identify probable future transition patterns rather than to provide parcel-level certainty (Gaur et al., 2023).

4.5.4 Comparative interpretation of forecast years

When the three forecast years are read together, they show a consistent and logically continuous sequence of future land transformation. The strongest common element in all maps is the persistent expansion and consolidation of built-up land. The built-up area remains centred on the existing urban core, but the maps also show scattered and corridor-like outward expansion into the agricultural fringe. This suggests that future urban growth in Gorakhpur is likely to be shaped by both core densification and peri-urban conversion.

A second common feature is the relative persistence of agricultural land as the dominant outer class, even in the forecast years. However, the maps clearly indicate that this agricultural dominance becomes less spatially continuous over time. Instead of appearing as a broad uninterrupted landscape, agricultural land becomes increasingly divided by built-up patches and development corridors. This means that the most important forecasted process is not total disappearance of agriculture, but its fragmentation and increasing encroachment by urban land.

The forest class remains confined to limited clusters, mainly toward the eastern side of the mapped area, while water bodies continue to appear in fixed low-lying positions. Their persistence in the forecast maps suggests that the model recognizes these as relatively stable landscape features. However, even when the total extent of water and forest does not visually collapse, their ecological and planning value may still be reduced if surrounding urbanization intensifies. This is especially relevant in Gorakhpur because the city lies in the Rapti–Rohini

basin and has a bowl-shaped geographical setting, making hydrological sensitivity an important planning concern (District Gorakhpur, 2026).

Overall, the forecasted results for 2035, 2045, and 2055 indicate that the future land system of Gorakhpur City is likely to remain on a strong urbanization path. All forecast maps, irrespective of model type, show the same central tendency: built-up land continues to dominate and expand, agricultural land remains extensive but increasingly fragmented, and forest and water persist mainly as localized patches. Because no forecast area-statistics table has been shared here, the present interpretation remains qualitative; however, the spatial message of the maps is very clear. The future of Gorakhpur appears to be one of continued urban expansion, increasing peri-urban conversion, and growing pressure on the non-urban landscape. This makes LULC forecasting especially valuable for forward-looking planning, which is one of the main purposes highlighted in the LULC modelling literature (Gaur et al., 2023).

4.6 Discussion

The results of the present study clearly indicate that Gorakhpur City has undergone substantial land use/land cover transformation, with the most dominant trend being the continuous expansion of built-up land and the corresponding reduction of agricultural land. The observed maps for 2005, 2015, and 2025 show that the city has gradually shifted from a relatively compact urban form toward a more extensive and spatially dominant built-up landscape. This trend is also supported by the area statistics, where built-up land increased from 42.8 km² in 2005 to 73.5 km² in 2025, while agricultural land declined from 145.6 km² to 118.3 km² during the same period. Similar results have been reported in earlier studies on Gorakhpur urban area, where rapid urban growth was found to be associated with decline in agricultural land and increasing land surface pressure (Tyagi and Sahoo, 2022).

A major point emerging from the predicted maps is that future urban growth in Gorakhpur appears to involve both edge expansion and urban infill. Edge expansion is visible in the spread of red built-up patches into the peri-urban fringe, whereas infill is reflected in the increasing continuity and densification of the already urbanized central area. This combination is significant because it suggests that the city is not only growing outward but also becoming more spatially consolidated internally. In practical terms, that means the future

urban form may become more dominant, more continuous, and more difficult to separate clearly from the surrounding peri-urban landscape. Similar urban-growth behavior has been documented in other Indian cities where remote sensing and GIS studies have shown that built-up expansion commonly occurs at the expense of agricultural land and open areas (Khan et al., 2019; Pandey & Seto, 2015).

The comparison between the ANN/MLP and MLP outputs also deserves discussion. Both model outputs show the same overall direction of change, namely continued urban expansion around the existing city core and progressive pressure on surrounding agricultural land. This agreement between the two model families is important because it strengthens confidence in the broad scenario. At the same time, the maps suggest slight local differences in change allocation, especially around the central waterbody and some mixed urban–non-urban transition zones. The ANN/MLP output appears somewhat more aggressive in central infill, while the MLP output preserves some water-linked features more distinctly. This kind of variation is expected in predictive modelling because different model structures often produce similar broad futures while differing in local transition allocation (Gaur et al., 2023). The results further suggest that forest, wasteland, and water bodies did not experience changes as dramatic as those observed in built-up and agricultural land, but this does not mean that they remained unaffected. Forest showed only a slight overall decline, and water bodies also recorded relatively small numerical change. However, the spatial maps indicate that these classes became increasingly enclosed within a more dominant urban matrix.

This issue is particularly serious in the case of Gorakhpur because the city has a highly sensitive physical setting. The official district profile states that Gorakhpur is situated in the basin of the Rapti and Rohini rivers and has a bowl-shaped geographical form, while flood-risk materials on Gorakhpur note that the Rapti–Rohini basin is recurrently flood-prone and hydrologically vulnerable (District Gorakhpur, 2026; NIDM, 2013). In such a setting, continued expansion of built-up and impervious surfaces can intensify runoff concentration, reduce infiltration opportunity, and increase the pressure on low-lying urban land and drainage systems. Therefore, the forecast maps should be discussed not only as urban-growth outputs, but also as indicators of possible future **environmental stress** if land conversion continues without planning control.

The present findings have important implications for urban planning and environmental

management. First, the peri-urban agricultural belt appears to be the most vulnerable land category and should therefore be treated as a strategic planning zone. Second, the persistence of water bodies and forest patches in the forecast maps suggests that future planning should focus not only on maintaining their areal extent but also on protecting their surrounding ecological buffers and connectivity. Third, because the city is geographically flood-sensitive, urban growth should be guided through a water-sensitive, risk-aware, and environmentally informed planning approach rather than uncontrolled fringe development. Previous work on Gorakhpur urbanization has also emphasized that unregulated urban sprawl can intensify land pressure, natural-resource stress, and sustainability challenges (Urbanization and Urban Sprawl in Gorakhpur City, 2024).

Overall, the discussion of the historical and forecasted results shows that Gorakhpur City is undergoing a long-term transformation from an agriculture-dominated landscape to a more built-up and spatially consolidated urban system. The observed increase in built-up land, the decline and fragmentation of agricultural land, and the continued future pressure indicated by the predictive maps all point toward the same conclusion: urban expansion is the dominant land transformation process in Gorakhpur. In a flood-prone and environmentally sensitive city such as Gorakhpur, this process has implications that go beyond simple land conversion. It directly affects ecological balance, land-use sustainability, water sensitivity, and future urban resilience. Therefore, the findings of this study strongly support the need for integrated and sustainable land-use planning in Gorakhpur City (District Gorakhpur, 2026; Gaur et al., 2023).

CHAPTER 5

CONCLUSIONS AND RECOMMENDATIONS

5.1 Conclusion

The study demonstrates significant LULC transformation in Gorakhpur city, characterized by rapid urban expansion and decline in agricultural land. The integration of GIS and deep learning techniques provided reliable prediction of future land-use scenarios. Multi-temporal satellite imagery was used to prepare LULC maps for 2005, 2015, and 2025, and these were analyzed to understand the spatial and temporal pattern of land transformation. The study found that Gorakhpur has experienced significant landscape change during the study period, with built-up land showing continuous expansion and agricultural land showing a steady decline. This indicates that urbanization has become the dominant force shaping the city's land system.

The results show that agricultural land remained the largest land-cover class in all observed years, but its overall area decreased over time. In contrast, built-up area increased strongly, reflecting rapid urban growth, infrastructural development, and expansion of the city into surrounding peri-urban zones. The maps also revealed that this transformation was not only numerical but spatial in nature. The city expanded outward from the existing core, while internal infill and consolidation also increased. As a result, Gorakhpur changed from a relatively compact urban settlement into a more extensive and spatially dominant urban landscape.

Although forest, wasteland, and water bodies did not show changes as large as built-up and agricultural land, these classes were also affected by the changing urban environment. Their spatial distribution suggests increasing fragmentation and pressure from surrounding development. This means that the environmental implications of land transformation extend beyond simple area change and include the weakening of ecological continuity and open-space balance.

The forecasting results for future years further suggest that the observed trend is likely to continue. The predicted maps indicate that built-up area may continue expanding in and around the present urban core, while agricultural land may remain under increasing pressure and become more fragmented in future decades. Therefore, the study concludes that Gorakhpur City is undergoing continuous and long-term urban transformation, and that this

process has direct implications for land-use sustainability, environmental quality, and future urban planning.

Overall, the study demonstrates that GIS, remote sensing, and forecasting techniques are effective tools for understanding past and present land transformation and for visualizing likely future scenarios. The findings provide a useful scientific basis for planning and policy decisions related to urban growth management in Gorakhpur City.

5.2 Recommendations

Based on the findings of the study, several recommendations may be proposed for future land-use planning and urban management in Gorakhpur City.

- First, urban growth should be guided through a more controlled and planned development strategy. Since built-up land is increasing rapidly, future urban expansion should be regulated through proper land-use zoning and development control. This would reduce random peri-urban sprawl and help direct growth toward suitable locations.
- Second, agricultural land in the peri-urban belt should be protected. The study clearly shows that agricultural land is the most vulnerable class under present urbanization trends. Because agricultural land contributes to food production, landscape balance, and open-space buffering, it should not be treated merely as reserve land for urban expansion.
- Third, water bodies and green spaces should be conserved more effectively. Even where their numerical decline is limited, increasing urban pressure around them may reduce their ecological value and environmental function. Therefore, buffer zones around water bodies, green patches, and environmentally sensitive areas should be maintained.
- Fourth, GIS and remote sensing should be used as regular planning tools by local authorities. Periodic updating of LULC maps can help identify land-conversion hotspots, monitor urban spread, and support timely intervention before land-use problems become severe.

Finally, forecast maps should be integrated into future planning exercises. Since the prediction results identify the probable direction of future urban growth, they can support

master planning, infrastructure design, environmental protection, and sustainable land-use regulation.

5.3 Limitations of the Study

Although the study provides useful results, certain limitations should be acknowledged.

- One limitation relates to the use of medium-resolution satellite imagery. Such imagery is suitable for city-scale and regional analysis, but it may not capture very fine land-cover details, especially in mixed urban fringe areas where built-up land, open land, and vegetation occur in close proximity.
- Another limitation concerns possible confusion between classes with similar spectral characteristics. For example, built-up area and barren land, or vegetation and agricultural land, may show similar spectral responses under certain conditions. This can affect classification precision in some locations.
- The study is also limited by the availability of reference and validation data. Since the accuracy of supervised classification depends on the quality of training samples and validation points, some uncertainty is always present in the final classified outputs.
- The forecasting results should also be interpreted carefully. Future prediction maps are not exact representations of future reality; rather, they are probable scenarios based on historical trends, selected variables, and model assumptions. Therefore, they are most useful for understanding the likely direction of change rather than for making exact local predictions.

5.4 Future Scope

The present study offers several possibilities for future improvement and extension.

- Future work may use higher-resolution satellite imagery to improve class discrimination and capture finer land-cover details, especially in urban fringe areas. This would help produce more precise maps and better identify small-scale landscape transformations.
- Further studies may also include additional variables such as road networks, population growth, land value, infrastructure concentration, and flood-prone areas.

Integrating these factors would improve the realism and planning usefulness of future land-use prediction.

- Another important future scope is the integration of LULC forecasting with environmental assessment. In Gorakhpur, this may include linking land transformation with flood risk, drainage conditions, land surface temperature, or ecological stress. Such analysis would provide a more comprehensive understanding of the consequences of urban growth.
- Future research may also compare multiple forecasting approaches, such as CA-Markov, LSTM, CNN-LSTM, or hybrid deep learning models, in order to identify the most suitable method for urban growth prediction in the local context.

Lastly, the analysis may be extended to ward-level or neighborhood-level planning units so that the results become more directly useful for local governance, municipal planning, and sustainable urban management.

In conclusion, the study confirms that Gorakhpur City is undergoing rapid land transformation, with built-up expansion emerging as the dominant trend and agricultural land becoming the most vulnerable class. The use of GIS, remote sensing, and forecasting methods has made it possible to understand both historical change and future land-use tendencies. Therefore, the study provides a valuable base for sustainable land-use planning,
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PUBLICATIONS AND PRESENTATIONS

1-

Title: Land Use/Land Cover Transformation and Forecasting in Gorakhpur City Using GIS and Deep Learning Approaches

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2-

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